MILITARY SPECIFICATION

EJECTOR, PARACHUTE, CARTRIDGE ACTUATED, XM233 METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Ejector, Parachute, Cartridge Actuated, XM233 Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with Ballistically Deployed Troop Chest Reserve-Type Parachute System.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATION

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings

DRAWINGS

U. S. Army Munitions Command

F10536011	- Ejector, Parachute, Cartridge Actuated XM233, Metal Parts Assembly
C9215469	- Carton, Packing, Ammunition, for Ejector Parachute, Cartridge Actuated, XM233
C9215470	- Box, Packing, Ammunition for Ejector, Parachute, Cartridge Actuated, XM233
IEL-10536011	- Inspection Equipment List for Ejector, Parachute, XM233, Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1377

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2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply:

ASA B46.1 - Surface Texture

Application for copies should be addressed to the American National Standards Institute (ANSI) 1430 Broadway, New York, N. Y. 10018.

3. REQUIREMENTS

3.1 General - Materials, parts and assemblies shall comply with the requirements specified in the applicable drawings, referenced specifications.

3.2 Material - The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.3 Mechanical properties. - Mechanical properties required by the drawing shall be uniform throughout each part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.4 Hydrostatic pressure

3.4.1 The tubes, orifice and manifold shall withstand a constant internal pressure of 6000 pounds per square inch (psi), minimum, for 15 seconds, minimum and shall show no evidence of leakage or mechanical failure (see 6.2)

3.4.2 The initiator chamber and cap, shall withstand a constant internal pressure of 14,000 to 15,000 pounds psi for 15 seconds, minimum, and shall show no evidence of leakage or mechanical failure (see 6.2).

3.5 Firing pin release. - The firing pin of the initiator metal parts assembly, less cotter pin, shall release from the cocked position when the application of an axial load of 15 to 30 pounds has withdrawn the initiator pin .370 inches minimum and .425 pounds maximum.

3.6 Metal parts assembly. - Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled hand tight.

3.7 Protective finish. - Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

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3.8 Workmanship - The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects - All components shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.8.2 Burr - No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter - No part of assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions - Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection. - Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample -

4.2.1 Initial production sample. - At the beginning of regular production, a sample shall be submitted in accordance with contract requirements. The sample shall consist of 5 metal parts assemblies and 10 sets of unassembled components for visual examination, dimensional examination and tests. The samples shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test. - After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government Laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure. - Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Inspection provisions

4.3.1 Lot

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 ejectors maximum, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and criteria in Tables I and II. For an individual defect (single characteristic), the criteria in Table I shall be used. For total defects within a major or minor group, the criteria in Table II shall be used. All nonconforming material shall be rejected. Inspection for "O" rings shall be in accordance with MIL-STD-413. The roughness comparison specimens prescribed in ASA-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter.

TABLE I

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

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4.3.2 (Cont'd)

TABLE II

Total Allowable Defects per Sample Size

Sample Sizes for Major Defects	110	150	225	300
No. of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over - 43	6	8	11	14
Sample Sizes for Minor Defects	**110**	**150**	**225**	**300**
No. of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over- 48	8	11	17	20

4.3.2.1 Tube, Inner, Drawing D10536010 covering a detail of Drawing F10536011.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Pitch diameter of thread, max.	Gage
102.	Minor diameter of thread, max.	Gage
103.	Concentricity of "O" ring to outside shoulder Dia.	Gage
104.	Large inside diameter	Gage
105.	Diameter of "O" ring groove	Gage
106.	"O" ring groove width	Gage
107.	"O" ring groove wall thickness	Gage
108.	Concentricity of "O" ring groove	Gage
109.	Concentricity of large outside diameter to small outside diameter	Gage
110.	Depth of inside chamfer, threaded end	Gage
111.	Small outside diameter	Gage
112.	Large outside diameter	Gage

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4.3.2.1 (Cont'd)

	Defect	Method of Inspection
MAJOR (Cont'd)		
113.	Chamfer diameter, threaded end	Gage
114.	Angle of chamfers, threaded end	Gage
115.	Width of outside chamfer, threaded end	Gage
116.	Length of large outside diameter	Gage
117.	Length from small end to thread	Gage
118.	Effective length of thread	Gage
119.	Protective finish inadequate or defective	Visual
120.	Surface finish, small outside diameter, incorrect	Visual
121.	Surface finish of inside chamfers, incorrect	Visual
122.	Burrs or sharp corners	Visual
123.	Metal defective	Visual
124.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, min.	Gage
202.	Total length	Gage
203.	Length of large inside diameter	Gage
204.	Profile of "O" ring groove configurations	Gage
205.	Radius or chamfers incorrect, unless otherwise specified	Gage
206.	Radius or chamfers missing	Visual
207.	Part number missing, incorrect or illegible	Visual
208.	Surface finish incorrect	Visual

4.3.2.2 Pin, Seating, Drawing C10536013 covering a detail of Drawing F10536011.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter thread, min	Gage
102.	Major diameter thread, min.	Gage
103.	Metal defective	Visual
104.	Burr	Visual
105.	Foreign matter	Visual
106.	Protective finish inadequate or defective	Visual

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Pitch diameter of thread, max.	Gage
202.	Large outside diameter	Gage
203.	Small outside diameter	Gage
204.	Length of thread relief	Gage
205.	Diameter of thread undercut	Gage
206.	Contour of tip	Gage
207.	Total length	Gage
208.	Length from tip to large diameter	Gage
209.	Width of large diameter	Gage
210.	Radius or chamfer incorrect	Gage
211.	Radius or chamfer missing	Visual
212.	Part number missing, incorrect or illegible	Visual
213.	Surface finish incorrect	Visual

4.3.2.3 Plug , Manifold, Drawing B10536014 covering a detail of Drawing F10536011.

CRITICAL

None defined.

MAJOR

101.	Pitch diameter thread, min.	Gage
102.	Major diameter thread, min.	Gage
103.	Metal defective	Visual
104.	Burr	Visual
105.	Foreign matter	Visual

MINOR

201.	Pitch diameter of thread, max.	Gage
202.	Diameter of thread undercut	Gage
203.	Total length	Gage
204.	Length of thread undercut	Gage
205.	Head thickness	Gage
206.	Depth of hole	Gage
207.	Outside head diameter	Gage
208.	Distance across flats	Gage
209.	Spherical diameter, outside flats	Gage
210.	Width of flat on head diameter	Gage
211.	Radius or chamfer, incorrect	Gage
212.	Radius or chamfer, missing	Visual
213.	Surface finish incorrect	Visual

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4.3.2.4 Inner Tube Assembly C10536017 covering a detail of F10536011

	Defect	Method of Inspection
CRITICAL		
	None defined.	
MAJOR		
101.	Snug fit of cushion	Manual
MINOR		
	None defined.	

4.3.2.5 Manifold, Drawing F10536015 covering a detail of Drawing F10536011.

CRITICAL		
	None defined.	
MAJOR		
101.	Pitch diameter of large thread, max.	Gage
102.	Minor diameter of large thread, max.	Gage
103.	Pitch diameter of medium threads, max. (2 places)	Gage
104.	Minor diameter of medium threads, max. (2 places)	Gage
105.	Pitch diameter of small thread, max.	Gage
106.	Minor diameter of small thread, max.	Gage
107.	True position of small diameter hole at bottom of large threaded hole to small diameter hole	Gage
108.	True position of medium thread pitch diameter, to pitch diameter of large thread.	Gage
109.	Perpendicularity of bottom of large threaded cavity to pitch diameter of large thread	Gage
110.	Perpendicularity of outside surface to pitch diameter of large threads	Gage
111.	True position of hole in bottom of large threaded hole to pitch diameter of large threads	Gage
112.	Large inside diameter	Gage
113.	Chamfer diameter, small threaded end	Gage
114.	Depth of chamfer, small threaded end	Gage
115.	Chamfer diameter, medium threaded end	Gage
116.	Width of large thread relief	Gage
117.	Inside diameter preceeding medium threads, (2 places)	Gage

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4.3.2.5 (Cont'd)

	Defect	Method of Inspection
118.	Inside diameter following medium threads, (2 places)	Gage
119.	Surface finish of inside chamfers incorrect	Visual
120.	Burrs or sharp corners	Visual
121.	Metal defective	Visual
122.	Foreign matter	Visual

MINOR

201.	Effective thread length, medium threads (2 places)	Gage
202.	Effective thread length, small thread	Gage
203.	Location of threaded holes	Gage
204.	Hole depth of small internal holes	Gage
205.	Depth of threaded holes	Gage
206.	Depth to medium threads (2 places)	Gage
207.	Thread length medium threads (2 places)	Gage
208.	Diameter of small hole at bottom of large thread hole	Gage
209.	Width of center portion	Gage
210.	Total height	Gage
211.	Length from small thread face to shoulder	Gage
212.	Diameter of small hole at bottom of small thread hole	Gage
213.	Outside configuration dimensions	Gage
214.	Total length	Gage
215.	Radius or chamfers incorrect	Gage
216.	Radius or chamfers missing	Visual
217.	Part number missing, incorrect or illegible	Visual
218.	Surface finish incorrect	Visual

4.3.2.6 Orifice, Drawing C10536016 covering a detail of Drawing F10536011.

CRITICAL

None defined.

MAJOR

101.	Pitch diameter external thread, min.	Gage
102.	Major diameter external thread, min.	Gage
103.	Pitch diameter internal thread, max.	Gage
104.	Minor diameter internal thread, max.	Gage
105.	Small inside diameter	Gage
106.	Length of effective internal thread	Gage

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4.3.2.6 (Cont'd)

	Defect	Method of Inspection
107.	Diameter of counterbore	Gage
108.	Length of inside diameter, small end	Gage
109.	Diameter of undercut, external thread	Gage
110.	Perpendicularity of shoulder to pitch diameter of external thread	Gage
111.	Surface finish of internal chamfers, large end, incorrect	Visual
112.	Protective finish inadequate or defective	Visual
113.	Metal defective	Visual
114.	Burrs or sharp corners	Visual
115.	Foreign matter	Visual

MINOR

201.	Pitch diameter external thread, max.	Gage
202.	Pitch diameter internal thread, min.	Gage
203.	Parallelism of large end to external thread shoulder	Gage
204.	True position of internal pitch diameter to external pitch diameter	Gage
205.	Depth of counterbore	Gage
206.	Chamfer diameter, large end	Gage
207.	Width of external thread relief	Gage
208.	Internal thread length, max.	Gage
209.	Large inside diameter	Gage
210.	Length from large end to external thread shoulder	Gage
211.	Length from large end to small inside diameter	Gage
212.	Total length	Gage
213.	Distance across flats	Gage
214.	Inside diameter, small end	Gage
215.	Radius or chamfers incorrect	Gage
216.	Radius or chamfers missing	Visual
217.	Surface finish, incorrect	Visual
218.	Part number missing, incorrect or illegible	Visual

4.3.2.7 Tube, Outer, Drawing D10536018 covering a detail of Drawing F10536011.

CRITICAL

None defined.

MAJOR

101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Small inside diameter	Gage

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4.3.2.7 (Cont'd)

	Defect	Method of Inspection
104.	Large inside diameter	Gage
105.	True position of large inside diameter to small inside diameter	Gage
106.	Concentricity of "O" ring groove to small inside diameter	Gage
107.	Perpendicularity of threaded end to large inside diameter	Gage
108.	Inside diameter of "O" ring groove	Gage
109.	"O" ring groove width	Gage
110.	"O" ring groove wall thickness	Gage
111.	Length of small inside diameter	Gage
112.	Chamfer width, max, threaded end	Gage
113.	Angle of chamfers, threaded end	Gage
114.	Outside diameter, max.	Gage
115.	Outside diameter preceeding thread	Gage
116.	Wall thickness, min.	Gage
117.	Length from large end to thread	Gage
118.	Effective length of thread	Gage
119.	Protective finish inadequate or defective	Visual
120.	Surface finish of large inside diameter, incorrect	Visual
121.	Burrs or sharp corners	Visual
122.	Metal defective	Visual
123.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, max.	Gage
202.	Total length	Gage
203.	Profile of "O" ring configurations	Gage
204.	Radius chamfers incorrect, unless otherwise specified	Gage
205.	Radius or chamfers missing	Visual
206.	Part number missing, incorrect or illegible	Visual
207.	Surface finish incorrect	Visual

4.3.2.8 Chamber, Initiator, Drawing D10520407, covering a detail of Drawing F10536011.

	Defect	Method of Inspection
CRITICAL		
	None defined.	

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4.3.2.8 (Cont'd)

	Defect	Method of Inspection
MAJOR		
101.	Pitch diameter of large external thread, min.	Gage
102.	Major diameter of large external thread, min.	Gage
103.	Pitch diameter of small external thread, min.	Gage
104.	Major diameter of small external thread, min	Gage
105.	Pitch diameter of internal thread, max.	Gage
106.	Minor diameter of internal thread, max.	Gage
107.	Concentricity of datum location with pitch diameter of small external thread	Gage
108.	Concentricity of pitch diameter of large thread with cavity	Gage
109.	Concentricity of large external thread pilot with large external thread.	Gage
110.	Concentricity of counterbore with cavity, including perpendicularity with base of counterbore	Gage
111.	Diameter of counterbore	Gage
112.	Diameter of large thread pilot	Gage
113.	Depth of cavity	Gage
114.	Length to datum location	Gage
115.	Thickness of wall, min.	Gage
116.	Metal defective	Visual
117.	Protective finish, inadequate or defective	Visual
118.	Surface finish of cavity and angle of gas port improper	Visual
119.	Burr or sharp corner	Visual
120.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of large external thread, max.	Gage
202.	Pitch diameter of small external thread, max.	Gage
203.	Pitch diameter of internal thread, min.	Gage
204.	Diameter of small external thread relief	Gage
205.	Diameter of small external thread pilot	Gage
206.	Diameter of hole in gas port	Gage
207.	Diameter of cavity	Gage
208.	Diameter of chamfer at internal thread	Gage
209.	Depth of counterbore	Gage
210.	Depth of threaded hole	Gage

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4.3.2.8 (Cont'd)

	Defect	Method of Inspection
211.	Length of large external thread	Gage
212.	Length of gas port	Gage
213.	Length of small thread relief	Gage
214.	Length from face of gas port to small thread	Gage
215.	Length of external thread pilot	Gage
216.	Total length	Gage
217.	Thickness of wall, max.	Gage
218.	Marking missing, incorrect or illegible	Visual
219.	Radius or chamfer missing or incorrect	Visual
220.	Surface finish improper unless otherwise classified	Visual

4.3.2.9 Cap, Initiator, Drawing F10535978, covering a detail of Drawing F10536011.

CRITICAL

None defined.

MAJOR

101.	Pitch diameter of small internal thread, max.	Gage
102.	Minor diameter of small internal thread, max.	Gage
103.	Pitch diameter of large internal thread, max.	Gage
104.	Minor diameter of large internal thread, max.	Gage
105.	Concentricity of pitch diameter of large internal thread (with midsection cavity, including perpendicularity with base of large cavity.)	Gage
106.	Concentricity of pitch diameter of small internal thread with midsection cavity including perpendicularity with base of large cavity counterbore	Gage
107.	Concentricity of hole between small cavity and boss cavity with midsection cavity, including perpendicularity with base of large cavity counterbore	Gage
108.	Concentricity of undercut in large cavity with large internal thread	Gage
109.	Concentricity of boss cavity with hole between small cavity and boss cavity.	Gage
110.	Concentricity of flange with hole between small cavity and boss cavity	Gage

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4.3.2.9 (Cont'd)

	Defect	Method of Inspection
111.	Concentricity of body, large end with large internal thread	Gage
112.	Concentricity of small cavity with midsection cavity	Gage
113.	Concentricity of thread cavity with large internal thread	Gage
114.	Diameter of hole between boss cavity and small cavity	Gage
115.	Diameter of boss cavity	Gage
116.	Diameter of large midsection cavity	Gage
117.	Diameter of small midsection cavity	Gage
118.	Diameter of undercut	Gage
119.	Length of midsection cavity effective thread, min.	Gage
120.	Thread damaged or not full	Visual
121.	Metal defective	Visual
122.	Protective finish inadequate or defective	Visual
123.	Surface finish of thread undercut improper	Visual
124.	Burr	Visual
125.	Foreign matter	Visual

MINOR

201.	Pitch diameter small internal thread, min.	Gage
202.	Pitch diameter large internal thread, min.	Gage
203.	Pitch diameter set screw thread	Gage
204.	Minor diameter set screw thread	Gage
205.	Diameter of chamfer at base of large threaded cavity counterbore	Gage
206.	Diameter of boss flange	Gage
207.	Diameter of body	Gage
208.	Diameter of cotter pin holes	Gage
209.	Diameter of counterbore at base of large threaded cavity	Gage
210.	Depth from chamfer at base of large threaded cavity counterbore to base of midsection cavity	Gage
211.	Depth of large threaded cavity counterbore	Gage
212.	Depth to base of large threaded cavity counterbore	Gage
213.	Depth around internal boss	Gage
214.	Depth from base of counterbore of large threaded cavity to chamfer at base of large midsection cavity	Gage

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4.3.2.9 (Cont'd)

	Defect	Method of Inspection
215.	Total length	Gage
216.	Length of flats at small end	Gage
217.	Length across flats at small end	Gage
218.	Length of flange at small end	Gage
219.	Length of head	Gage
220.	Length from large threaded cavity counterbore to boss in small end	Gage
221.	Width of counterbore base of large threaded cavity	Gage
222.	Location of set screw hole	Gage
223.	Location of cotter pin hole	Gage
224.	Radius or chamfer missing or incorrect	Visual
225.	Surface finish improper, except as otherwise classified	Visual

4.3.2.10 Pin, Initiator, Drawing C11730531, covering a detail of Drawing F10536011.

CRITICAL

None defined.

MAJOR

101.	Concentricity of groove in head with body	Gage
102.	Concentricity of head with body	Gage
103.	Pitch diameter, max.	Gage
104.	Diameter of body	Gage
105.	Diameter of head (front)	Gage
106.	Diameter of head (rear)	Gage
107.	Diameter of groove in body	Gage
108.	Diameter of groove in head	Gage
109.	Diameter of pin hole	Gage
110.	Width of head groove	Gage
111.	Width of slot in head	Gage
112.	Location of pin hole from rear head	Gage
113.	Location of pin hole from front head	Gage
114.	Location of body groove	Gage
115.	Thickness of head (rear)	Gage
116.	Hardness	Gage
117.	Metal defective	Visual
118.	Protective finish inadequate	Visual

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4.3.2.10 (Cont'd)

	Defect	Method of Inspection
119.	Surface finish improper	Visual/Gage
120.	Burr	Visual
121.	Foreign matter	Visual

MINOR

201.	Pitch diameter min.	Gage
202.	Minor diameter, max.	Gage
203.	Width across head flats	Gage
204.	Total length	Gage
205.	Radius of body groove	Visual
206.	Radii of chamfer missing or incorrect	Visual
207.	Surface finish improper except as otherwise classified	Visual

4.3.2.11 Screw Pin, Drawing C11730532, covering a detail of Drawing F10536011

CRITICAL

None defined

MAJOR

101.	Pitch diameter, min.	Gage
102.	Locking element missing or incorrect	Visual/Gage

MINOR

201.	Pitch diameter, max.	Gage
202.	Major diameter, max.	Gage
203.	Diameter of head	Gage
204.	Length of body	Gage
205.	Thickness of head	Gage
206.	Depth of slot	Gage
207.	Width of slot	Gage
208.	Profile of head	Visual
209.	Radius or chamfer missing or incorrect	Visual
210.	Protective finish inadequate	Visual
211.	Surface finish improper	

4.3.2.12 Seal, Drawing B10520415 covering a detail of Drawing F10536011.

	Defect	Method of Inspection

CRITICAL

None defined

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4.3.2.12 (Cont'd)

	Defect	Method of Inspection
MAJOR		
101.	Diameter, outside, min.	Gage
102.	Width, min.	Gage
103.	Thickness, min.	Gage
104.	Cut, cracked or split	Visual
105.	Flash	Visual
106.	Foreign matter	Visual
MINOR		
201.	Diameter, outside, max.	Gage
202.	Width, max.	Gage
203.	Thickness, max.	Gage

4.3.2.13 Housing, Firing Pin Drawing C10520410 covering a detail of Drawing F10536011.

CRITICAL		
1.	Diameter of striker hole, min.	Gage
MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of striker hole with cavity	Gage
104.	Concentricity of cavity with pitch diameter of thread	Gage
105.	Concentricity of body with pitch diameter of thread	Gage
106.	Perpendicularity of face with body	Gage
107.	Diameter of body, min.	Gage
108.	Diameter of cavity	Gage
109.	Diameter of flange, min.	Gage
110.	Diameter of boss	Gage
111.	Diameter of undercut, min.	Gage
112.	Total length, max.	Gage
113.	Width of undercut	Gage
114.	Thickness of striker hole	Gage
115.	Location of thread	Gage
116.	Chamfer at striker hole	Gage
117.	Height of boss	Gage

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4.3.2.13 (Cont'd)

	Defect	Method of Inspection
118.	Metal defective	Visual
119.	Protective finish, inadequate or defective	Visual
120.	Surface finish of cavity, improper	Visual
121.	Burr	Visual
122.	Foreign matter	Visual

MINOR

201.	Pitch diameter of thread, max.	Gage
202.	Diameter of body, max.	Gage
203.	Diameter of striker hole, max.	Gage
204.	Diameter of flange, max.	Gage
205.	Diameter of wrench holes	Gage
206.	Depth of wrench holes	Gage
207.	Total length, min.	Gage
208.	Length of body to flange	Gage
209.	Location of wrench holes	Gage
210.	Marking missing, incorrect or illegible	Visual
211.	Radius or chamfer, missing or incorrect, except as otherwise classified	Visual
212.	Surface finish improper, except as otherwise classified	Visual

4.3.2.14 Pin, Firing, Drawing C10520411 covering a detail of Drawing F10536011

CRITICAL

None defined.

MAJOR

101.	Concentricity of large cavity with body	Gage
102.	Concentricity of striker with body	Gage
103.	Concentricity of cavity with body	Gage
104.	Symmetry of pin relief holes with small cavity	Gage
105.	Symmetry of ball hole counterbore with ball hole	Gage
106.	Diameter of striker	Gage
107.	Diameter of body	Gage
108.	Diameter of ball hole, max.	Gage
109.	Diameter of ball hole, counterbore	Gage
110.	Diameter of pin relief holes, min.	Gage
111.	Depth of ball counterbore, min.	Gage

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4.3.2.14 (Cont'd)

	Defect	Method of Inspection
112.	Length of striker, min.	Gage
113.	Location of ball hole, counterbore	Gage
114.	Metal defective	Visual
115.	Protective finish inadequate or defective	Visual
116.	Surface finish of body improper	Visual
117.	Burr or sharp corner	Visual
118.	Foreign matter	Visual

MINOR

201.	Diameter of large cavity	Gage
202.	Diameter of small cavity	Gage
203.	Diameter of ball hole, min.	Gage
204.	Diameter of pin relief holes, max.	Gage
205.	Depth of large cavity	Gage
206.	Depth to bottom of cavity	Gage
207.	Depth of ball hole counterbore, max.	Gage
208.	Length of striker, max.	Gage
209.	Total length	Gage
210.	Location of pin relief holes	Gage
211.	Marking missing incorrect or illegible	Visual
212.	Radius or chamfer missing or incorrect	Visual
213.	Surface finish improper, except as otherwise classified	Visual

4.3.2.15 Spring, Initiator, Drawing B10520412, covering a detail of Drawing F10536011.

CRITICAL

None defined

MAJOR

101.	Load at specified height	Gage
102.	Closed length, max.	Gage
103.	Ends incorrect	Visual
104.	Protective finish inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual

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4.3.2.15 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Diameter, outside	Gage
202.	Free length	Gage

4.3.2.16 Filter, Drawing C10520414, covering a detail of drawing F10536011.

CRITICAL

None defined.

MAJOR

101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of thread with center hole and outside diameter	Gage
104.	Symmetry of holes with outside diameter	Gage
105.	Diameter of center hole	Gage
106.	Diameter of holes	Gage
107.	Depth of center hole	Gage
108.	Depth of slot, max.	Gage
109.	Location of holes	Gage
110.	Angular location of holes	Gage
111.	Nylon patch not properly located or fused	Manual or Visual
112.	Number of holes incorrect	Visual
113.	Metal defective	Visual
114.	Protective finish inadequate or defective	Visual
115.	Burr	Visual
116.	Foreign matter	Visual

MINOR

201.	Pitch diameter of thread, max.	Gage
202.	Diameter of undercut	Gage
203.	Diameter of body	Gage
204.	Depth of slot, min.	Gage
205.	Total length	Gage
206.	Length of body	Gage
207.	Width of undercut, max.	Gage
208.	Width of slot	Gage
209.	Location of slot	Gage
210.	Marking missing, incorrect or illegible	Visual
211.	Chamfer missing or incorrect	Visual
212.	Surface finish improper	Visual

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4.3.2.17 Initiator Assembly Drawing F11731643 less all drawings except C10520410, C10520411 and MS19060-S.

CRITICAL	Defect	Method of Inspection
1.	Firing pin protrusion	Gage
2.	Steel ball missing	Gage
MAJOR		
101.	Burr	Visual
102.	Foreign matter	Visual
MINOR		
	None defined.	

4.3.2.18 Initiator Assembly, Drawing F11731643 covering a detail of Drawing F10536011.

CRITICAL		
	None defined.	
MAJOR		
101.	Firing pin not cocked	Manual
102.	Initiator pin not free to rotate	Manual
103.	Any part missing, except as otherwise classified	Visual
104.	Improper assembly	Visual
105.	Cotter pin improperly inserted	Visual
106.	O-ring not lubricated	Visual
107.	Marking missing, incorrect or illegible	Visual
MINOR		
201.	Shipping cap missing	Visual
202.	Protective finish inadequate or defective	Visual

4.3.2.19 Assembly, Drawing F10536011

CRITICAL	
	None defined

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4.3.2.19 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Lot number or part number missing, incorrect or illegible	Visual
102. Metal defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
201. Shipping plug missing	Visual
202. Protective finish inadequate or defective	Visual

4.3.2.20 Packing in Carton, Ammunition, Drawing C9215469.

CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Contents loose	Manual
202. Cut or damaged	Visual
203. Marking, missing, incorrect or illegible	Visual

4.3.2.21 Packing in Wooden Box, Drawing C9215470.

CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual and Manual

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4.3.2.21 (Cont'd)

	Defect	Method of Inspection
203.	Strapping missing, loose, broken or improperly assembled	Visual and Manual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of units incorrect	Visual

4.3.3 Inspection testing - The following tests shall be performed in accordance with the provisions of 4.4:

4.3.3.1 Hardness - Fifty units from each heat treated lot of components shall be tested to determine compliance with the requirements noted on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.3.2 Hydrostatic Pressure

4.3.3.2.1 Each tube and manifold shall be subjected to an internal hydrostatic pressure test. Failure of any unit to comply with the requirements of 3.4 shall be cause for rejection of the component.

4.3.3.2.2 Each chamber and cap subassembly shall be subjected to the hydrostatic pressure test. Failure of any subassembly to comply with the requirements of 3.4 shall be cause for rejection of the subassembly.

4.3.3.3 Firing pin release - The initiator pin of each partial assembly consisting of all parts except the cotter pin and chamber shall be subjected to the firing pin release test. Failure of any partial assembly to comply with the requirements of 3.5 shall be cause for rejection of the subassembly.

4.3.3.4 Protective finish. - The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standards listed on the drawing.

4.3.4 Inspection equipment - The examination and tests shall be made using equipment listed on IEL-10536011.

4.4 Test methods and procedures

4.4.1 Hardness. - Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the hardness requirements specified on the detail drawing.

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4.4.2 Hydrostatic pressure. - Prior to the application of protective finish, the tubes and manifold individually, shall be mounted in a test device, using the fixture for sealing all openings, and subjected to an internal hydrostatic pressure in accordance with the requirements of 3.4. The testing device used shall internally seal the tubes and manifold, at the prescribed seal surface, without applying any axial load other than that applied by the liquid pressure.

4.4.2.1 Hydrostatic pressure - Prior to the application of protective finish, the chamber and cap subassembly shall be mounted in the device, using the fixture for properly sealing all openings, and subjected to the hydrostatic pressure to determine compliance with the requirements of 3.4. The test device used shall internally seal the components at the prescribed seal surfaces without applying an axial load other than that applied by the liquid pressure.

4.4.3 Firing pin release - The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

4.4.4 Protective finish. - Test procedures for protective finish shall be in accordance with the applicable documents referenced on the component drawing.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A - Packing and marking shall be in accordance with Drawings C9215469 and C9215470.

6. NOTES

6.1 Ordering data - Procurement documents should specify the title, number, and date of this specification.

6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A485

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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